Graphs 4 — Quadratic graphs · Solving $ax^2 + bx + c = 0$ graphically

Squared paper needed. Plot the points carefully and join them with a *smooth curve* — never straight lines. Marks in brackets. ★ = harder question.

1 Copy and complete the table, then draw the graph of $y = x^2 - 2x - 3$ for $-2 \le x \le 4$. [6]

x	-2	-1	0	1	2	3	4
y		0		-4		0	

Use your graph to write down the solutions of $x^2 - 2x - 3 = 0$.

2 **a** Copy and complete the table, then draw the graph of $y = 2x^2 + x - 3$ for $-3 \le x \le 2$. [8]

x	-3	-2	-1	0	1	2
$2x^2$		8		0	2	
$+x$		-2		0	1	
-3	-3	-3	-3	-3	-3	-3
y		3		-3	0	

b Use your graph to solve the equation $2x^2 + x - 3 = 0$.

3 The profit p (€ millions) of a start-up company t years after launch is modelled by $p = 5t - kt^2$ for $0 \leq t \leq 10$, where k is a constant. [11]

t	0	2	4	6	8	10
p	0		12			0

a Use the table to show that $k = 0.5$, then copy and complete the table.

b Draw the graph of p against t.

c Use the graph to estimate **(i)** the greatest profit and when it occurs, **(ii)** the period during which the profit is at least €8 million.

Total: 25 marks

Check: a quadratic graph is a parabola — U if $a > 0$, ∩ if $a < 0$ — and symmetric · the solutions of $ax^2 + bx + c = 0$ are where the curve crosses the x-axis (two, one or none) · read the greatest/least value at the turning point.